

Enhancing Financial Literacy Through Narrative and Puzzles

A serious game approach for young adults

MASTER DISSERTATION

Ana Catarina Antunes Santos

MASTER IN INFORMATICS ENGINEERING



UNIVERSIDADE da MADEIRA

A Nossa Universidade

www.uma.pt

December | 2024

Enhancing Financial Literacy Through Narrative and Puzzles

A serious game approach for young adults

MASTER DISSERTATION

Ana Catarina Antunes Santos

MASTER IN INFORMATICS ENGINEERING

SUPERVISOR

Mara Sofia Gomes Dionísio

CO-SUPERVISOR

Pedro Filipe Pereira Campos



FACULTY OF EXACT SCIENCES AND ENGINEERING

MASTER OF SCIENCE DEGREE IN INFORMATICS ENGINEERING

**ENHANCING FINANCIAL LITERACY
THROUGH NARRATIVE AND PUZZLES:
A SERIOUS GAME APPROACH FOR YOUNG ADULTS**

Ana Catarina Antunes Santos

Orientado por:

Mara Sofia Gomes Dionísio

Pedro Filipe Pereira Campos

Constituição do júri de provas públicas:

Prof.a Dr.a Karolina Baras (Professora Auxiliar da Universidade da Madeira), Presidente

Dr.a Sandra Olim (Investigadora de Pós-Doutoramento do ITI/LARSyS), Vogal

Prof.a Dr.a Mara Dionísio (Professora Auxiliar da Universidade da Madeira), Vogal

Tuesday 11th March, 2025

Resumo

A literacia financeira é crucial para o bem-estar económico, tanto a nível pessoal como nacional, especialmente na sociedade complexa e em rápida evolução. Apesar da sua importância, persiste um défice significativo no conhecimento financeiro básico, mesmo entre indivíduos com níveis elevados de educação. Este défice é particularmente preocupante entre os jovens adultos, pois impede-os de tomar decisões informadas, gerir eficazmente as suas finanças e assegurar um futuro estável.

Esta tese explora o desenho, desenvolvimento e avaliação de dois jogos sérios destinados a melhorar a literacia financeira em jovens adultos, com particular ênfase na poupança e no planeamento financeiro. O primeiro jogo, DreamScape, é um jogo interativo guiado por uma narrativa que inclui puzzles financeiros, enquanto o segundo, Where's the Finance: Goldcrest Curse, adota o formato de um jogo passivo.

O documento apresenta uma visão abrangente do desenho conceptual, da metodologia de desenvolvimento e da implementação destes protótipos digitais. Além disso, detalha um rigoroso processo de avaliação que analisou a usabilidade, os resultados de aprendizagem, o envolvimento dos utilizadores e a atratividade geral dos jogos. Os resultados da avaliação demonstraram que o DreamScape ofereceu uma experiência altamente imersiva graças ao seu desenho baseado em narrativa, envolvendo os jogadores profundamente no jogo. Por outro lado, o Where's the Finance foi mais eficaz na apresentação de tópicos de literacia financeira, com os participantes a demonstrarem uma compreensão mais clara e uma maior retenção dos conceitos. Ambos os jogos receberam feedback positivo em relação à acessibilidade e relevância, destacando o seu potencial como ferramentas educativas.

Os resultados sugerem que os jogos sérios são uma ferramenta promissora para melhorar a literacia financeira entre jovens adultos, especialmente quando equilibram conteúdos educativos com mecânicas envolventes. Esta investigação destaca o potencial da aprendizagem baseada em jogos para preencher défices educacionais críticos e recomenda uma exploração mais aprofundada da aplicação de jogos sérios para o ensino de outras competências essenciais para a vida.

Keywords: Literacia Financeira; Jovens Adultos; Jogos Sérios; Educação; Comportamento Financeiro

Abstract

Financial literacy is crucial for both personal and national economic well-being, particularly in today's complex and rapidly evolving society. Despite its importance, a significant gap in basic financial knowledge persists, even among individuals with higher education. This gap is especially concerning for young adults, as it prevents them from making informed decisions, managing finances effectively, and securing a stable future.

This thesis explores the design, development, and evaluation of two serious games aimed at improving financial literacy in young adults, with a particular emphasis on savings and budgeting. The first game, DreamScape, is an interactive narrative-driven game featuring financial puzzles, while the second, Where's the Finance: Goldcrest Curse, adopts the idle game format.

The document provides a comprehensive overview of these digital prototypes' conceptual design, development methodology, and implementation. Additionally, it details a rigorous evaluation process that assessed usability, learning outcomes, user engagement, and overall appeal. Evaluation results demonstrated that DreamScape offered a highly immersive experience through its narrative-driven design, engaging players deeply in the gameplay. In contrast, Where's the Finance was more effective at presenting financial literacy topics, with participants demonstrating clearer understanding and improved retention of key concepts. Both games received positive feedback for accessibility and relevance, highlighting their potential as educational tools.

The findings suggest that serious games are a promising tool for enhancing financial literacy among young adults, particularly when designed to balance educational content with engaging mechanics. This research highlights the potential of game-based learning in addressing critical educational gaps and recommends further exploration into the application of serious games for teaching other essential life skills.

Keywords: Financial Literacy; Young Adults; Serious Games; Education; Financial Behaviour

Acknowledgments

I would like to express my gratitude to my advisors, Dr. Mara Dionísio and Dr. Pedro Campos, for their support and guidance throughout the development of this thesis. I am especially thankful to Dr. Mara Dionísio for her dedication, availability, and the valuable insights she provided during this process.

I am deeply thankful to Dr. Paulo Bala for his essential contributions to the completion of this work. His consistent support and expertise were greatly appreciated throughout this journey.

I would like to extend my gratitude to the docents of the University of Madeira, who played a significant role in my academic journey during this master's degree, and to my colleagues from the eGames Lab for their encouragement and support. In particular, I am grateful to Constança Freitas for her collaboration on this thesis.

I would also like to thank the Portuguese Recovery and Resilience Program (PRR), IAP-MEI/ANI/FCT under Agenda no.26, C645022399-00000057 (eGamesLab) for funding this research.

A special thank you to my dear friend Sara Trafaria for her willingness to assist with prototype testing, which was crucial for evaluating outcomes and improving the project.

Finally, I am deeply thankful to my family, my mother, brother, and grandmother, for their constant support, encouragement, and belief in me. Their presence has been a source of strength throughout my academic journey. Thank you for always being there for me.

Table of Contents

List of Figures	viii
List of Tables.....	xi
1 Introduction	1
1.1 Thesis Context	2
1.2 Objectives & Research Questions.....	3
1.3 Thesis Contributions	4
1.4 Thesis Structure	5
2 Literature Review	7
2.1 Financial Literacy	7
2.2 Serious Games	8
2.3 Serious Games for Financial Literacy	10
2.4 Interactive Narratives & Narrative Game Design.....	13
2.5 Research Opportunities	14
2.6 Development Technologies	15
3 Initial Research & Prototype.....	19
3.1 Exploratory Study	19
3.1.1 Protocol & Metrics	20
3.1.2 Results.....	21
3.1.3 Discussion	22
3.2 Preliminary Prototypes.....	23
3.2.1 Low-Fidelity Prototype (Figma).....	24
3.2.2 Prototype (Figma) Usability Testing	25
3.2.3 Protocol & Measures	26
3.2.4 Sample	26
3.2.5 Results	27
3.3 Main Takeaways from Preliminary Prototypes of the DreamScape	28
3.4 Prototype Gameplay	28
4 Design & Modeling.....	32

4.1	Prototype 1: <i>DreamScape</i>	32
4.1.1	Game Narrative	32
4.1.2	Game Mechanics	35
4.1.3	High Fidelity Prototype	40
4.2	DreamScape Software Modeling	42
4.2.1	System Requirements	42
4.2.2	Functional Requirements	42
4.2.3	Non-Functional Requirements	44
4.2.4	Use Cases	47
4.3	Prototype 2: <i>Where's the Finance: Goldcrest Curse</i>	49
4.3.1	Game Narrative & Game Mechanics	49
4.3.2	High Fidelity Prototype	50
4.3.3	Software Modeling	51
4.3.4	Functional Requirements	51
4.3.5	Non-Functional Requirements	54
4.3.6	Use Cases	55
5	Implementation	57
5.1	Prototype 1: DreamScape	57
5.1.1	Yarn Spinner	58
5.1.2	Sleep Sequence Mini-Games	59
5.1.3	Budgeting Methods Mini-Games	63
5.2	Protytpe 2: Where's the Finance: Goldcrest Curse	69
5.2.1	Core Gameplay Mechanics	72
6	Evaluation	80
6.1	Prototype 1: DreamScape	80
6.1.1	Protocol & Measures	80
6.2	Data Analysis	81
6.2.1	Sample	83
6.3	Quantitative Data	83
6.4	Qualitative Data	87
6.5	Prototype 2: Where's the Finance?: Goldcrest Curse	89
6.5.1	Protocol & Metrics	89

6.5.2 Quantitative Data	90
6.6 Qualitative Data	90
7 Discussion	92
7.1 Limitations & Future Work	94
8 Conclusion	97
References	100

PREVIEW

List of Figures

1	The figure 1 illustrates the iterative design process, beginning with a workshop to define key financial topics, followed by the creation of a paper prototype and a Figma prototype.	19
2	Paper prototype. Screen 1 (Top Left): Showcases the elements present in the game, including the characters (a girl and a boy) available for the user to choose from. The gargoyle will later ask questions. Screen 2 (Top Right): The gargoyle asks the user questions related to FL, engaging the player in the learning process. Screen 3 (Bottom): The room opens after the user answers all questions correctly, allowing the user to exit the house and complete the scenario.....	24
3	Figma prototype. Screen 1 (Top Left): The dark hall with a lamp at the centre requires the insertion of three symbols to proceed. Screen 2 (Top Right): The characters' room, where three hidden symbols representing Savings, Needs, and Wants must be found. Screen 3 (Bottom Left): The kitchen scenario at night, where the user tries to drink water but finds none, triggering a mini-game. Screen 4 (Bottom Right): The mini-game screen, where the user must allocate the correct percentage of water into buckets labelled Wants, Needs, and Savings, reflects the 50/30/20 budgeting rule. This screen includes a description explaining the objective of the mini-game to the user.	25
4	2D scenario of the messages	32
5	2D scenario of the Character Heading Home.....	33
6	3D scenario of the room	33
7	Player finding the first Object	34
8	2D scenario of the player going to sleep	34
9	Sleep sequence puzzle mini games	35
10	Introduction of the 50-30-20 budgeting Method	36
11	Introduction of the Envelop Budgeting Method	36
12	Introduction of the zero-based Budgeting Method	36
13	Visual of the 50-30-20 Budgeting method mini-game.....	38
14	Visual of the envelop Budgeting method mini-game	39

15	Visual of the zero-based Budgeting method mini-game	39
16	Use cases overview	48
17	Global view of the prototype	57
18	Demonstration of the initial node of yarn spinner 3D room scene.	58
19	Example of the connection between the Yarn script and the CSharp script.	58
20	Flowchart of the first sleep sequence mini-game	59
21	Method responsible for setting the initial state of the puzzle.	60
22	Method responsible for "Shuffle" the puzzle pieces.	60
23	Flowchart of the second sleep sequence mini-game	60
24	Method responsible for loading a collection of sprites from a folder	61
25	Method responsible for defining how many cards (buttons) will there be in the mini-game	61
26	Method responsible for populating the list of buttons based on the number of buttons ..	61
27	Flowchart of the third sleep sequence mini-game	62
28	Method responsible for handling the interaction between items and the object used to catch the items.....	62
29	Methods responsible for managing the player score by increasing or decreasing it based on the prototype events	63
30	Definition of the class "ItemSpawner", which is responsible for controlling the spawning of items at regular intervals in the prototype.	63
31	Flowchart of the 50-30-20 budgeting method mini-game	64
32	OnBeginDrag and OnDrag methods, where the OnBeginDrag method disables raycasting to allow the object to be moved, and the OnDrag method updates the object position based on the user input, ensuring smooth dragging	64
33	OnEndDrag method, which checks if the dragged object is dropped on a valid target.	65
34	UpdateDropStatus method, which tracks and updates the number of correct drops	65
35	CalculateBudgetDistribution method, which calculates and displays the allocation of income into the categories of needs, wants, and savings based on the 50-30-20 rule.	65
36	Flowchart of the envelop budgeting method mini-game.....	66
37	DistributeBudget method, illustrating how the user income is parsed, divided into equal parts, and displayed for distribution across categories.	66
38	Flowchart of the zero-based budgeting method mini-game	67

39	OnSubmitButtonClicked method	68
40	Continuation of the OnSubmitButtonClicked	68
41	Simulation method	69
42	Overview of the flow structure for the prototype	69
43	overview of the flow structure for the tutorial scene.....	70
44	overview of the flow structure for the tutorial scene.....	71
45	overview of the flow structure for the tutorial scene.....	71
46	overview of the flow structure for the tutorial scene.....	71
47	Method "MoveCamera"	72
48	Method "ZoomCamera"	72
49	Method "ClampCamera"	73
50	Method "Awake"	73
51	Method "Start"	74
52	Method "OnCharacterClicked"	74
53	Method "OnInstaObjectsClicked"	75
54	Method "OnMouseDown"	75
55	Method "OnPointerClick"	76
56	Method "AddPoints"	76
57	Method "DeducPoints"	76
58	Method "MoveArrow"	77
59	Method "AskForHelp"	77
60	Method "Yes"	78
61	Method "No"	78
62	Method "ShowHelpCircleOnInstaObject"	78

List of Tables

1	Games tested, Name, Positive Aspects, and Negative Aspects	12
2	Comparison of Game Engines	16
3	Comparison of Narrative Tools	17
4	Wilcoxon test statistics - Self-Assessment	84
5	Wilcoxon test statistics - Financial Literacy	84
6	PXI Questionnaire Median & Standard Deviation by Construct	85
7	Spearman's Correlation between Constructs	86
8	Quantitative Findings Summary	91
9	Qualitative Data Thematics Summary	91

Lista de Acrónimos

2D Two Dimensional

3D Three Dimensional

AI Artificial Intelligence

FL Financial Literacy

FR Functional Requirements

NFR Non-Functional Requirements

PXI Player Experience Inventory

SPSS Statistical Package for the Social Sciences

UC Use Cases

UI User Interface

1 Introduction

Financial literacy (FL) is widely recognized as a crucial determinant of individual financial well-being; studies consistently highlight the strong correlation between low levels of FL and unfavourable financial outcomes, including the increase of financial stress, inadequate saving and budgeting practices [1, 2]. These challenges underscore the importance of preparing young adults with essential financial skills to make informed decisions and secure their future prosperity. The significance of FL extends beyond personal benefits, influencing broader economic stability and social well-being, as emphasized by initiatives in several countries to prioritize financial education [3].

Despite its importance, many individuals, particularly young adults, continue to struggle with low levels of FL, even in environments where education levels are generally high. According to a study by Scheresberg [4], young adults with lower FL are more likely to rely on costly borrowing methods, such as payday loans or high-interest credit cards, and are less likely to maintain emergency savings and actively prepare for their retirement. Young adults face increasingly complex financial landscapes shaped by digital technologies, innovative financial products, and evolving service ecosystems [5]. To address these challenges, nations such as New Zealand and Portugal have launched comprehensive FL strategies. New Zealand's 2008 framework, for example, coordinates initiatives to bridge educational gaps, while Portugal's 2021–2025 strategy emphasizes targeted financial education campaigns [3, 6].

However, traditional FL interventions, including didactic games, educational presentations, e-learning courses, and quizzes, often fail to actively engage learners or sustain their interest [7, 8]. These methods tend to neglect critical elements such as exploration, strategic thinking, and creativity, which are essential for meaningful learning experiences [9]. Rasco et al. [10] emphasize the limited availability of FL programs designed specifically for young adults while pointing out an even more significant lack of innovative, game-based approaches that utilize interactivity and engagement to enhance FL skills.

This thesis addresses these gaps by exploring the potential of narrative-driven serious games as tools for teaching FL to young adults. Serious games, games designed with educational purposes in mind, have shown promise in knowledge transfer and skill development across various domains [11, 12]. Despite their demonstrated effectiveness, their application to FL education, particularly

through narrative and puzzle-based formats, remains underexplored. Building on insights from serious games research, this thesis examines two narrative-driven games developed as part of the eGames Lab project [13]: DreamScape and Where’s the Finance: Goldcrest Curse.

DreamScape, a computer-based game developed and designed within the scope of this thesis, integrates an immersive storyline with puzzle mechanics to simulate real-life financial decisions. By solving these puzzles, players engage with FL concepts in an educational and interactive manner.

In contrast, Where’s the Finance: Goldcrest Curse, designed by Constança Freitas as part of her Master’s Degree in Interactive Media Design (MDMI), adopts a web-based idle game format. This approach offers a gradual, narrative-driven learning experience, emphasizing accessibility and flexibility. While the two games differ in structure and delivery, they share a common goal: to create engaging, effective tools for enhancing FL among young adults.

This collaboration will allow for the investigation of how serious games can enhance FL among young adults between the ages of 18 and 35, addressing key questions about the role of narrative, interactivity, and game mechanics in fostering learning. It also evaluates the relative strengths and limitations of different formats, computer-based and web-based, in achieving educational objectives. The findings contribute to the body of work on serious games and explore the potential of using innovative tools for financial education.

1.1 Thesis Context

This research was funded by the Portuguese Recovery and Resilience Program (PRR), IAP-MEI/ANI/FCT under Agenda no.26, C645022399-00000057 (eGamesLab) [13], a collaborative initiative to foster innovation and competitiveness in Portugal game development sector. My contributions within this framework focused on developing two narrative-driven serious games, DreamScape and Where’s the Finance: Goldcrest Curse, to enhance FL among young adults.

While the eGames Lab offers shared resources and a collaborative environment, my work focused primarily on the technical implementation of the game prototypes; this involved coding, implementing, and prototyping the mechanics for both DreamScape and Where’s the Finance: Goldcrest Curse.

For DreamScape, I was responsible for both the design and technical implementation of the game. The game mechanics were developed by integrating puzzle-based elements to simulate real-life

financial decisions while creating an immersive narrative experience. In addition to the gameplay mechanics, I also wrote the narrative elements, including the progression of the game and the intertwining of financial decisions with the storyline. The narrative structure of DreamScape was designed to engage players in a story that intertwines personal finance decisions with the gameplay. For the visual design, Constança contributed significantly by designing most of the two-dimensional (2D) art assets used in the game. Her artwork was essential in bringing the narrative to life, helping to create a visually cohesive experience that complemented the storyline. As for the three-dimensional (3D) assets, most were sourced from the Unity Asset Store, while we created a few custom assets using Blender to fit specific gameplay needs.

For *Where's the Finance: Goldcrest Curse*, Constança led the game's design, including the narrative structure and progression. She crafted the storyline, which unfolds gradually as players engage with the game, making it an accessible and interactive way to learn financial concepts. Constança also designed all the 2D art assets and provided the functional and non-functional requirements for the game. My role was to gather the requirements from Constança and implement them into a working prototype, focusing on translating her design and requirements into a playable web-based serious game.

1.2 Objectives & Research Questions

The main objective of this research is to assess the effectiveness of two distinct narrative-driven serious games as educational tools for enhancing FL among young adults. This thesis evaluates how each serious game, DreamScape and *Where's the Finance: Goldcrest Curse*, can engage players and promote knowledge retention in the context of FL education. By focusing on these two games, the research explores whether narrative and interactive mechanics can offer effective, engaging solutions for teaching FL.

A key goal of this study is to evaluate DreamScape, a computer-based serious game that integrates an immersive narrative with puzzle-solving mechanics designed to help players apply FL concepts in real-world scenarios. Through this evaluation, the research will investigate how the game's combination of story-driven content and problem-solving challenges contributes to both player engagement and the retention of FL knowledge.

The second objective is to evaluate Where’s the Finance: Goldcrest Curse, a serious web-based game designed by my colleague Constança Freitas. The game follows a narrative idle format, gradually enabling players to build their FL. The objective is to assess how this format, which is characterized by relaxed gameplay, still effectively engages players and helps them learn about financial concepts at a manageable pace.

Ultimately, this research will assess how both games engage young adults, whether through interactive puzzles or incremental learning and how well these formats support long-term retention and application of FL skills. The findings will offer insights into the design elements contributing to narrative-driven serious games’ effectiveness in FL education.

The following research goals guided this thesis:

RG1: Creating a serious game that educates young adults on financial topics, specifically budgeting and savings, to improve their FL and decision-making skills.

RG2: Assessing the effectiveness of the game mechanics in promoting financial education among young adults, specifically regarding knowledge retention and engagement with the topics of budgeting and savings.

This research addresses the following research questions:

RQ1: Which game mechanics can be effective for teaching financial topics within the serious game context?

RQ2: What gamification elements within the serious game can potentially engage players in the learning experience?

1.3 Thesis Contributions

This thesis contributes to the field of FL education through the technical development, implementation, and evaluation of two serious game prototypes: DreamScape and Where’s the Finance: Goldcrest Curse. These contributions cover programming, system integration, and game mechanics design, each aimed at creating engaging, narrative-driven experiences to support FL learning.

In the web-based idle game Where’s the Finance: Goldcrest Curse, designed by Constança Freitas, our primary focus was on implementing the game according to her design and functional

specifications. Her narrative-driven design was translated into a playable prototype, coding the core idle mechanics to facilitate an accessible, gradual approach to financial learning.

This thesis also makes a scientific contribution through two publications developed within the scope of this dissertation:

Article 1 - **Santos, A.**, Freitas, C., Bala, P., Campos, P., Dionisio, M. (2024). Financial DreamScape: Puzzle Narrative Games for Financial Education. ACM International Conference on Interactive Media Experiences, 4, 421-425. <https://dl.acm.org/doi/10.1145/3639701.3663647> [14]

Article 2 - Freitas, C., **Santos, A.**, Bala, P., Campos, P., Dionisio, M. (2024). Where's The Finance?: A Transmedia Storytelling Experience to Engage Young Adults in Financial Educational Content. International Conference on Interactive Digital Storytelling. Springer, 2024, pp. 32-46. https://doi.org/10.1007/978-3-031-78450-7_3 [15]

1.4 Thesis Structure

The structure of this document has been organized to provide a logical understanding of the development, design, and implementation of both prototypes, DreamScape and Where's the Finance?: Goldcrest Curse, as well as the evaluations conducted throughout this process.

Chapter 2 is dedicated to the literature review, where key topics relevant to the thesis are explored within its context. This includes an analysis of serious games for FL, interactive narratives, and a review of the technologies essential for the development of the prototypes.

Chapter 3 presents the exploratory study conducted prior to the development of the prototypes, which informed the selection of topics. It also outlines the preliminary prototypes, describing their development process, evaluation, and the key takeaways from these prototypes.

Chapter 4 covers the design and modelling of both prototypes, including system requirements and software modelling.

Chapter 5 details the implementation of both prototypes, outlining the flow of each level and the features incorporated into the games.

Chapter 6 describes the evaluation of both prototypes, providing an analysis of the quantitative and qualitative data gathered.

Chapter 7 presents the discussion of this thesis, revisiting the research questions and explaining how they were addressed. This chapter also discusses the project's limitations and suggests future work directions.

Chapter 8 concludes this thesis by revisiting the gap in financial knowledge, examining how this research addresses it, and highlighting the key findings and their significance.

PREVIEW

2 Literature Review

This section presents the literature review conducted for this thesis, providing an overview of key topics relevant to the research. The review begins by exploring the concept of FL, emphasizing its importance and the critical need to address it within the young adult demographic. It then examines the role of serious games as an educational tool, focusing on their potential to engage learners and convey complex concepts effectively. Next, the intersection of serious games and FL is discussed, with examples of existing FL games that illustrate their application. Finally, the review explores interactive narratives and narrative game design, highlighting their significance in creating engaging and meaningful learning experiences within the context of game-based education.

2.1 Financial Literacy

The literature consistently underscores the critical role of FL in achieving positive financial outcomes. Low levels of financial knowledge are strongly linked to poor financial decision-making, as highlighted by Younis et al., Gezmen et al., Rasco et al., and Lusardi et al. [1, 2, 10, 16–18]. Despite its importance, Rasco et al. [10] note that relatively few FL programs are tailored for young adults, and even fewer employ engaging and sustainable methods to effectively improve financial knowledge.

In Portugal, efforts to enhance FL have gained progress, led by initiatives from Banco de Portugal. In 2008, the "Bank Customer Portal" was launched to distribute information on banking products and services, creating a foundation for broader educational efforts. By 2011, the National Financial Education Plan was introduced under the initiative "Todos Contam," offering general guidelines to promote FL across diverse audiences. This initiative expanded further in 2014 with the introduction of a teacher training program and instructional manuals specifically designed to facilitate financial education in schools [6, 19]. Most recently, the 2021–2025 FL plan broadened its scope to include schools, workplaces, local communities, and small to medium-sized enterprises. This comprehensive effort also integrates youth-focused awareness campaigns and partnerships with traditional media, addressing new challenges posed by digital financial products and post-pandemic economic disruptions. These efforts aim to counteract issues like impulsive decision-making and excessive borrowing, which have been aggravated by the growing prevalence of digital channels [6].

The need for FL is becoming increasingly critical in today’s fast-evolving economic landscape. Young adults, in particular, face a more complex financial environment than previous generations, shaped by digital technologies and the rise of innovative financial products. As Hayei et al. [5] highlight, younger individuals must navigate these complexities to make informed financial decisions, underscoring the importance of equipping them with the necessary skills and knowledge. However, FL remains underrepresented in formal educational learning plans, leaving many to rely on trial-and-error approaches. Cultural and social barriers, such as the perception of financial topics as taboo, further restrict effective financial education [20, 21].

This issue is particularly pronounced among young adults. According to Amari et al., [22], a significant portion of young adults lack an understanding of basic financial and economic concepts, leaving them unprepared for the financial decisions they face. Mitchell et al. [23] also emphasize that financial decisions must be confronted earlier in life, and without essential financial knowledge, mistakes made in youth can have costly and long-lasting consequences. This further reinforces the need for targeted financial education initiatives aimed at this demographic, to help them navigate these challenges and make informed financial choices.

Despite these challenges, the importance of FL extends beyond individual well-being to encompass national economic health. FL initiatives, such as those implemented by Banco de Portugal, play a vital role in addressing this gap. However, one promising approach is using serious games and gamification to teach FL. These methods combine entertainment with education, making complex financial concepts more accessible and engaging for young adults.

2.2 Serious Games

Video games are often associated with entertainment, particularly in nowadays society, where many people engage with them for leisure. However, their potential as powerful learning tools is increasingly recognized by educators and game developers [24]. Research suggests that active, hands-on engagement enhances learning effectiveness, and video games align well with this principle by encouraging practical, interactive activities that facilitate experiential learning [25].

Serious games, games designed as simulations of real-world scenarios to solve problems, have emerged as highly effective educational tools. They combine entertainment with educational goals, offering learners engaging and interactive platforms for acquiring knowledge. Zyda et al. [11] de-

scribe serious games as visual, interactive experiences that promote deep learning through challenges and problem-solving. Similarly, Bellotti et al. [26] highlight their ability to foster active learning, requiring players to dynamically respond to various scenarios.

One of the key strengths of serious games is their ability to motivate learners by integrating game mechanics such as rewards, leaderboards, and challenges to create immersive environments that encourage players to explore, strategize, and solve problems. This motivational aspect, as noted by Westera et al. [27], plays a critical role in shaping educational outcomes. Schrader et al. [28] further underscore the relevance of serious games in digital and distance education contexts, where maintaining engagement is often challenging.

The versatility of serious games is another factor contributing to their popularity in education. Anastasiadis et al. [29] discuss how gamification incorporating game elements into non-game settings enhances the learning experience by rewarding specific behaviours or achieving educational goals. Laamarti et al. [30] note that the increasing volume of research on serious games over the past decade reflects their growing adoption as an effective educational strategy.

Despite their success, the classification of serious games remains a topic of discussion. Connolly et al. [12] argue that, unlike entertainment games, serious games do not rely heavily on genres, as their primary focus is educational outcomes. Laporte et al. [31] add that the term "serious games" is often sufficient to describe games aimed at learning, regardless of their specific classification.

In practical applications, serious games thrive at simulating real-life scenarios and promoting "learning by doing". For example, Allal-Chérif et al. [32] highlight their use in corporate environments to train financial advisors, fostering essential qualities like flexibility, integrity, and reliability. This hands-on, experiential approach aligns with the broader principle that active participation significantly enhances knowledge retention and skill development.

Zhonggen et al. [33] identified one of the key reasons for the effectiveness of serious games in education as their ability to positively influence learners' moods, with players reporting higher levels of happiness when engaging with serious game-assisted learning. They also noted that these games, along with other educational technologies such as mobile applications, have been shown to improve academic achievements and encourage greater participation in learning activities.

The immersive nature of serious games is another significant factor contributing to their effectiveness. Cheng et al. [34] found that gaming immersion positively correlates with improved performance in science learning. Supporting this, a meta-analysis by Lamb et al. [35], which reviewed 46 empirical studies, highlighted that serious games not only help learners acquire cognitive abilities but also enhance their positive affect towards learning.

In the context of computer science education, Hakulinen et al. [36] explored the use of four card-based serious games to teach data structures and algorithms. Their findings suggest that such games can enhance student engagement and facilitate the understanding of complex concepts. Furthermore, they noted that most educational serious games are computer-based due to their ease of distribution. However, they highlighted limitations in non-computer games, which may lack the adaptability and interactivity of digital counterparts.

Hauge et al. [37] examined the application of serious games in engineering education, where they have been utilized to train both master-level students and industry professionals. These games demonstrated the ability to help players apply theoretical knowledge and develop collaboration skills. However, the effectiveness of these games depends on factors such as the groups' background knowledge and openness to gameplay. Homogeneous groups, those with similar expertise levels, background knowledge or characteristics, appeared to benefit more than inhomogeneous ones, as these serious games required collaboration to be successfully completed.

2.3 Serious Games for Financial Literacy

Cole et al. [2] emphasize the critical role of FL (FL) in fostering personal growth and economic development. Initiatives like FinCraft, an open-source gaming platform aimed at enhancing FL among youth through gamification, illustrate innovative approaches to addressing this need [10]. While FinCraft seeks to bridge the gap between financial knowledge and interest—an essential aspect of youth development—it lacks thorough user evaluation, leaving its impact on engagement and learning outcomes uncertain.

Although efforts such as FinCraft [10] and national education plans have achieved significant progress [6], Rasco et al. [10, 38] highlight the necessity of ongoing innovation to keep pace with the rapidly evolving financial landscape. In this context, the serious game developed for this thesis introduces a novel approach by incorporating a puzzle-based narrative. This design actively assesses